Quantum Hierarchical Risk Parity and Its Classical Counterpart

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Abstract Traditional portfolio optimization methods like Markowitz's mean-variance optimization often fall short due to instability, concentration, and sensitivity to estimation errors – particularly during market crises when correlations drastically increase and diversification benefits vanish. Hierarchical Risk Parity (HRP) addresses some of these issues by leveraging hierarchical clustering to improve diversification and portfolio stability. Meanwhile, recent advancements in quantum machine learning (QML) suggest powerful new approaches to modeling complex financial relationships through quantum feature mapping and density matrix techniques. This paper introduces two novel approaches: Quantum Hierarchical Risk Parity (QHRP) and Kernel-based Hierarchical Risk Parity (KHRP). QHRP integrates the strengths of HRP with quantum computing by encoding financial data into quantum density matrices and utilizing Frobenius distances for clustering to capture deeper asset relationships. Meanwhile, KHRP, which can be considered a classical version of QHRP, leverages kernel methods, specifically Maximum Mean Discrepancy (MMD) with a Gaussian kernel, to enhance asset relationship modeling. The real-world empirical analyses demonstrate that these methods consistently achieve superior risk-adjusted performance in comparison with traditional methods, paving the way for more robust and effective portfolio management strategies.

Keywords: Portfolio Optimization, Hierarchical Risk Parity, Quantum Machine Learning, Quantum Feature Maps, Density Matrices, Frobenius Distance

1 Introduction

There is an amusing yet insightful story often retold in finance circles about a mathematician who drowned in a stream with an average depth of just two inches (Elton et al, 2014). The moral is clear: relying solely on averages, volatilities, or correlations can be dangerously misleading. Just as an average depth does not adequately describe the variability of a stream, relying on single-dimensional measures like correlation is insufficient to capture the complex, often hidden, risks and relationships within financial markets. Moreover, correlations are not

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Supporting Material:
Reproducible Results: github.com/RiskLabAI/Notebooks.py/tree/main/optimization/quantum_hrp

static — they change significantly over time and tend to spike precisely during crises when diversification is most urgently needed (Buraschi et al, 2014). Ironically, investors frequently place their money in funds advertising diversification benefits based on correlations observed during normal market periods. Yet, these correlations often evaporate exactly when investors rely on them the most during market downturns. Indeed, Elton et al (1976) further emphasized that, despite the elegance of Markowitz's portfolio theory, practical implementation has been severely limited due to challenges in accurately estimating correlation matrices and related complexities.

Marcos Lopez de Prado offers a parallel lesson in modern finance: "*Backtesting is not a research tool. Feature importance is*"(De Prado, 2018). Before assessing feature importance, one must ensure their model can handle the complexity and dimensionality inherent in real-world data. While traditional methods like Markowitz mean-variance optimization have been cornerstone concepts since 1952 (Markowitz, 1952), their practical performance often suffers due to issues like instability, concentration, and sensitivity to estimation errors (Black and Litterman, 1992; Ban et al, 2018). In addition, due to their limited modeling framework, the number of features they take is usually limited and model-hardcoded. To overcome these practical challenges, researchers have begun exploring new techniques, significantly broadening the landscape of financial optimization methodologies. For instance, Hierarchical Risk Parity (HRP), introduced by Lopez de Prado (2016), cleverly bypasses many of the pitfalls associated with classical covariance-based methods by leveraging graph theory and clustering techniques, delivering more stable and diversified portfolios (Burggraf, 2021). Despite its advantages, the classical HRP suffers from the same limitations as it takes a limited and fixed number of features – to be more precise, one feature: correlation.

Quantum computing approaches promise to tackle previously intractable computational problems within finance. Recent developments like the Variational Quantum Eigensolver (VQE) and quantum annealing methods have demonstrated strong potential to optimize portfolios more efficiently than traditional approaches (Buonaiuto et al, 2023; Venturelli and Kondratyev, 2019). Quantum classifiers based on density matrices further showcase how quantum methods can elegantly encode complex relationships between assets (Garvin et al, 2025). And a quantum two-sample test that leverages the expressive power of a quantum feature map (Garvin et al, 2024) can improve the accuracy of the trading strategies graduation testing.

These quantum methods provide a fundamentally new lens through which we view the complexity of financial markets. As highlighted by recent reviews on quantum finance (Naik et al, 2025), integrating quantum algorithms into finance holds considerable promise for portfolio optimization, risk management, fraud detection, and derivative pricing, among other applications. Moreover, these advancements align closely with the broader goal of extracting richer information from data beyond simple measures like correlation.

This article proposes Quantum Hierarchical Risk Parity (QHRP), which combines the power of quantum computing and HRP to construct optimized portfolios. Not restricted by the limitations of classical covariance matrices, QHRP makes use of quantum feature mapping and Frobenius distances to capture high-dimensional, complex asset relationships. QHRP enhances portfolio robustness, stability, and performance via embedding assets into quantum states and measuring their similarity through quantum distance metrics. For a fair comparison, we additionally propose the classical version of the QHRP, Kernel-Based Hierarchical Risk Parity (KHRP), which uses Gaussian kernels to define a measure of similarity between assets. QHRP and its classical counterpart, KHRP, are not limited to any fixed number of constant features and provide users with the flexibility of having their own arbitrary set of features. This is a major advancement compared to the classical HRP method. We validate QHRP's practical efficacy through the real-world empirical evaluations. We demonstrate that quantum-enhanced methods can deliver superior financial outcomes when applied to large-scale real-world portfolios.

2 Quantum Hierarchical Risk Parity

This section outlines the proposed QHRP methodology, which integrates quantum machine learning techniques with the Hierarchical Risk Parity algorithm for optimal portfolio construction. The methodology is structured into five key stages: Quantum Feature Mapping, Density Matrix Construction, Frobenius Distance Calculation, Distance Matrix Construction, and Integration with HRP.

Algorithm 1 QHRP Algorithm

Require: Financial dataset $\mathcal{D} = \{\mathbf{x}_i^k \in \mathbb{R}^P\}_{i=1,\dots,N;\, k=1,\dots,T}$

Ensure: Portfolio weights $\mathbf{w} = \{w_1, \dots, w_N\}$

1. Quantum Feature Mapping:
  Encode each observation $\mathbf{x}_i^k$ into quantum feature maps $\varphi(\mathbf{x}_i^k)$ using unitary transformations $U(\mathbf{x}_i^k)$
  Construct average density matrices $\rho_i = \frac{1}{T}\sum_{k=1}^{T} |\mathbf{x}_i^k\rangle\langle\mathbf{x}_i^k|$ for each asset i
2. Frobenius Distance & Distance Matrix Calculation:
  Compute Frobenius distances $D_{i,j} = d_F(\rho_i, \rho_j) = \frac{1}{2}\sqrt{\mathrm{Tr}\,[(\rho_i - \rho_j)^2]}$ for all asset pairs (i, j) to form distance matrix D
3. Tree Clustering:
  Apply hierarchical clustering on D to build dendrogram $\mathcal{T}$ using minimum distance linkage
4. Quasi-Diagonalization:
  Rearrange asset order based on $\mathcal{T}$ to obtain quasi-diagonal covariance matrix Σ'
5. Recursive Bisection:
  Allocate portfolio weights $\mathbf{w}$ using recursive bisection aligned with hierarchical clusters
Return: Portfolio weights $\mathbf{w}$

2.1 Quantum Feature Mapping

Consider a financial dataset $\mathcal{D}$ comprising N assets, each with T observations. Each observation for asset i is represented as a feature vector $\mathbf{x}_i^k \in \mathbb{R}^P$, where $k \in \{1, \dots, T\}$ and P denotes the number of financial features. We define a quantum feature map $\varphi : \mathbb{R}^P \to \mathcal{H}$, where $\mathcal{H}$ is a Hilbert space realized on n qubits via a unitary transformation $U(\mathbf{x})$:

$$\varphi(\mathbf{x}) := U(\mathbf{x})|\mathbf{0}\rangle\langle\mathbf{0}|U(\mathbf{x})^\dagger \equiv |\mathbf{x}\rangle\langle\mathbf{x}|. \tag{1}$$

Here, $|\mathbf{x}\rangle = U(\mathbf{x})|\mathbf{0}\rangle$ is a pure quantum state encoding the feature vector $\mathbf{x}$, and $\varphi(\mathbf{x})$ is the corresponding density matrix. The feature map φ facilitates embedding classical financial feature vectors into a high-dimensional quantum state space, enabling the capture of complex interdependencies and correlations inherent in the data.

2.2 Density Matrix Construction

For each asset $i \in \{1, \dots, N\}$, we construct an average density matrix ρ_i by aggregating the quantum states of its T observations:

$$\rho_i = \frac{1}{T}\sum_{k=1}^{T} |\mathbf{x}_i^k\rangle\langle\mathbf{x}_i^k|. \tag{2}$$

Here, $\mathbf{x}_i^k$ denotes the k-th observation of asset i. The density matrix ρ_i represents an equally weighted statistical ensemble of the quantum-encoded observations for asset i, effectively capturing the aggregate quantum representation of the asset's behavior across all its observations. This aggregation allows for a comprehensive quantum characterization of each asset, which is essential for subsequent distance computations and portfolio optimization.

2.3 Frobenius Distance & Distance Matrix Calculation

To quantify the similarity between assets, we compute the Frobenius distance between their respective density matrices. The Frobenius distance d_F between two density matrices ρ_i and ρ_j is defined as:

$$d_F(\rho_i, \rho_j) = \frac{1}{2}\|\rho_i - \rho_j\| = \frac{1}{2}\sqrt{\mathrm{Tr}\left[(\rho_i - \rho_j)^2\right]}. \tag{3}$$

This metric measures the geometric distance between the quantum states of two assets, providing a basis for clustering in the subsequent steps. Using the Frobenius distances, we construct an $N \times N$ distance matrix $D = \{d_{i,j}\}_{i,j=1}^{N}$, where each entry $d_{i,j} = d_F(\rho_i, \rho_j)$ represents the distance between assets i and j. This matrix encapsulates the pairwise relationships between all assets based on their quantum-encoded representations.

2.4 Integration with Hierarchical Risk Parity

The distance matrix D is integrated into the HRP algorithm to perform optimal portfolio allocation. The HRP methodology consists of three stages: Tree Clustering, Quasi-Diagonalization, and Recursive Bisection.

Stage 1: Tree Clustering

1. **Hierarchical Clustering**: Apply hierarchical clustering on the distance matrix D to build a tree structure of asset clusters. Specifically, use agglomerative clustering to iteratively merge the closest pairs of assets or clusters based on the Frobenius distance.
2. **Linkage Criterion**: Define the distance between clusters using a linkage criterion (e.g., single linkage, complete linkage). For QHRP, we utilize the *minimum distance* (nearest point) linkage:
$$\tilde{d}_{i,u[k]} = \min_{j \in u[k]}\{d_{i,j}\}, \quad \forall i \notin u[k]. \tag{4}$$
3. **Cluster Formation**: Identify the pair of columns (i^*, j^*) such that
$$(i^*, j^*) = \underset{(i,j),\, i \neq j}{\arg\min} \left\{\tilde{d}_{i,j}\right\}, \tag{1}$$
and denote this cluster as $u[1]$.
4. **Distance Update**: Define the distance between the newly formed cluster $u[1]$ and each unclustered item i as
$$\dot{d}_{i,u[1]} = \min\left\{\tilde{d}_{i,j}\right\}_{j \in u[1]}. \tag{2}$$
Update the distance matrix D by appending $\dot{d}_{i,u[1]}$ and removing the clustered columns and rows $j \in u[1]$.
5. **Recursive Clustering**: Repeat steps 2 to 4 recursively, appending $N - 1$ such clusters to matrix D. The final cluster will contain all original items, and the clustering algorithm terminates.

Stage 2: Quasi-Diagonalization

Rearrange the order of assets based on the hierarchical clustering to achieve a quasi-diagonal covariance matrix. This ensures that similar assets are positioned adjacent to each other, enhancing the efficiency of the subsequent allocation process. The algorithm proceeds as follows:

1. For each row of the linkage matrix, merge the clusters recursively.
2. Replace clusters in $(y_{N-1,1}, y_{N-1,2})$ with their constituents until no clusters remain.
3. The output is a sorted list of original (unclustered) items, resulting in a quasi-diagonal covariance matrix where the largest values lie along the diagonal.

Stage 3: Recursive Bisection

Allocate portfolio weights using a top-down approach based on the quasi-diagonalized covariance matrix. The steps are as follows:

1. **Initialization**:

$$L = \{L_0\}, \quad \text{where } L_0 = \{1, \ldots, N\}, \tag{3}$$
$$w_n = 1, \quad \forall n \in \{1, \ldots, N\}. \tag{4}$$

2. **Recursive Allocation**: For each cluster $L_k \in L$ with $|L_k| > 1$:
   (a) **Bisection**: Split L_k into two subsets $L_k^{(1)}$ and $L_k^{(2)}$, where

$$|L_k^{(1)}| = \left\lfloor \frac{1}{2}|L_k| \right\rfloor, \tag{5}$$

   and the order is preserved.
   (b) **Variance Calculation**: Compute the variance of each subset using inverse-variance weighting:

$$\widetilde{V}_k^{(j)} = \widetilde{\mathbf{w}}_k^{(j)\top} V_k^{(j)} \widetilde{\mathbf{w}}_k^{(j)}, \quad j = 1, 2, \tag{6}$$

   where $V_k^{(j)}$ is the covariance matrix of subset $L_k^{(j)}$ and

$$\widetilde{\mathbf{w}}_k^{(j)} = \operatorname{diag}(V_k^{(j)})^{-1} \frac{1}{\operatorname{tr}\left[\operatorname{diag}(V_k^{(j)})^{-1}\right]}. \tag{7}$$

   (c) **Weight Allocation**: Determine the split factor α_k and allocate weights as follows:

$$\alpha_k = 1 - \frac{\widetilde{V}_k^{(1)}}{\widetilde{V}_k^{(1)} + \widetilde{V}_k^{(2)}}, \tag{8}$$
$$w_i \leftarrow \alpha_k \cdot w_i, \quad \forall i \in L_k^{(1)}, \tag{9}$$
$$w_i \leftarrow (1 - \alpha_k) \cdot w_i, \quad \forall i \in L_k^{(2)}. \tag{10}$$

3. **Termination**: Repeat the recursive allocation process until all clusters are reduced to single assets, i.e., $|L_k| = 1$ for all $L_k \in L$.

This concludes the first description of the HRP algorithm, which solves the allocation problem in deterministic logarithmic time, $T(n) = O(\log_2 n)$.

Inverse Variance Allocation

Stage 3 splits a weight in inverse proportion to the subset's variance. We can prove that such allocation is optimal when the covariance matrix is diagonal. Consider the standard quadratic optimization problem of size N:

$$\begin{aligned} \min_{\omega} \quad & \omega^\top V \omega \\ \text{subject to} \quad & \omega^\top a = 1_I, \end{aligned} \tag{11}$$

with solution

$$\omega = \frac{V^{-1}a}{a^\top V^{-1}a}. \tag{12}$$

For the characteristic vector $a = \mathbf{1}_N$, the solution is the minimum variance portfolio. If V is diagonal, then

$$\omega_n = \frac{V^{-1}_{n,n}}{\sum_{i=1}^N V^{-1}_{i,i}}. \tag{13}$$

3 Kernel-Based Hierarchical Risk Parity

We propose a KHRP framework, outlined in Algorithm 2, where asset distances are derived from a kernel function rather than classical correlation or quantum distance measures. Our approach uses a Gaussian kernel to capture nonlinear similarities among asset features, thereby quantifying differences via the Maximum Mean Discrepancy. We argue that the KHRP method constitutes a fair comparative benchmark for QHRP because both methods extend beyond traditional correlation analysis by capturing complex, nonlinear relationships in asset returns. In contrast to QHRP, which embeds features into a high-dimensional quantum state space, KHRP employs classical kernel techniques to measure asset dissimilarity, and both frameworks produce hierarchical clusterings that drive portfolio allocation.

We denote the set of assets by $\{1, 2, \dots, N\}$ and assume each asset i is represented by a feature matrix $\mathbf{X}_i \in \mathbb{R}^{T_i \times P}$, where T_i is the number of observations and P is the number of features. We define a Gaussian kernel between two feature vectors $\mathbf{x}$ and $\mathbf{y}$ as

$$k(\mathbf{x}, \mathbf{y}) \;=\; \exp\Bigl(-\frac{\|\mathbf{x} - \mathbf{y}\|^2}{2\sigma^2}\Bigr). \tag{14}$$

Here, $\sigma > 0$ is the kernel bandwidth parameter, $\|\mathbf{x} - \mathbf{y}\|$ denotes the Euclidean distance between $\mathbf{x}$ and $\mathbf{y}$, and $k(\mathbf{x}, \mathbf{y})$ represents their similarity. For two assets i and j with feature matrices $\mathbf{X}_i$ and $\mathbf{X}_j$ containing T_i and T_j observations respectively, we compute the MMD distance as:

$$\mathrm{MMD}(\mathbf{X}_i, \mathbf{X}_j)^2 \;=\; \frac{1}{T_i^2}\sum_{t=1}^{T_i}\sum_{s=1}^{T_i} k(\mathbf{x}_{i,t}, \mathbf{x}_{i,s}) + \frac{1}{T_j^2}\sum_{t=1}^{T_j}\sum_{s=1}^{T_j} k(\mathbf{x}_{j,t}, \mathbf{x}_{j,s}) - \frac{2}{T_i T_j}\sum_{t=1}^{T_i}\sum_{s=1}^{T_j} k(\mathbf{x}_{i,t}, \mathbf{x}_{j,s})\,. \tag{15}$$

In Equation 15, $\mathbf{x}_{i,t}$ is the t-th observation for asset i and $\mathbf{x}_{j,t}$ is the t-th observation for asset j; the sums compute intra-asset and cross-asset kernel similarities, and $\mathrm{MMD}(\mathbf{X}_i, \mathbf{X}_j)$ quantifies the distance between assets i and j. We form an $N \times N$ distance matrix $\mathbf{D}$ whose (i, j) entry is given by

$$\mathbf{D} \;=\; \bigl\{\mathrm{MMD}(\mathbf{X}_i, \mathbf{X}_j)\bigr\}_{i,j=1}^{N}. \tag{16}$$

Here, $\mathbf{D}$ encapsulates the pairwise MMD distances between all assets.

Following the computation of $\mathbf{D}$, we perform hierarchical clustering with linkage on the condensed distance matrix obtained from $\mathbf{D}$. The resulting dendrogram yields an ordering π of the assets. We then rearrange the covariance matrix $\boldsymbol{\Sigma}$ of asset returns according to π to form a quasi-diagonal matrix. Finally, we allocate portfolio weights using a recursive bisection method. For a subset S of assets, we compute inverse-variance weights as

$$\mathbf{w}_S \;=\; \frac{\boldsymbol{\sigma}_S^{-1}}{\sum_{i \in S} \frac{1}{\sigma_i}}\,, \tag{17}$$

where σ_i denotes the standard deviation of asset i and $\boldsymbol{\sigma}_S$ is the vector of standard deviations for assets in S. This recursive bisection proceeds until each cluster contains a single asset, resulting in the final portfolio weight vector $\mathbf{w} = \{w_1, \dots, w_N\}$.

Algorithm 2 KHRP Algorithm

Require: Financial dataset $\mathcal{D} = \{\mathbf{X}_i \in \mathbb{R}^{T_i \times P}\}_{i=1}^{N}$, asset return covariance matrix $\boldsymbol{\Sigma}$, and kernel bandwidth $\sigma > 0$
Ensure: Portfolio weights $\mathbf{w} = \{w_1, \ldots, w_N\}$
1: Compute pairwise MMD distances between asset features using Equation 15 to form the distance matrix $\mathbf{D}$ as in Equation 16.
2: Perform hierarchical clustering on the condensed form of $\mathbf{D}$ using a linkage method to obtain a dendrogram and an asset ordering π.
3: Rearrange the covariance matrix $\boldsymbol{\Sigma}$ according to π to obtain a quasi-diagonal covariance matrix $\boldsymbol{\Sigma}'$.
4: Allocate portfolio weights using recursive bisection with inverse-variance allocation as defined in Equation 17.
5: **return** The portfolio weights $\mathbf{w}$.

4 Numerical Experiments

In this section, we present the methodology for constructing the quantum feature map, the experimental setup, and the results of our numerical experiments. The analysis was conducted on a diversified portfolio of assets over the period from January 1, 2005, to January 1, 2025, as detailed in the provided source code and Appendix A.

4.1 Experimental Setup and Features

A core premise of this work is that QHRP and KHRP can leverage richer, multi-dimensional data to build superior portfolios. Our experimental setup, as defined in the implementation script, sets the number of features per asset to $P = 6$. While the feature generation function was external to the provided script, we describe here a representative set of features consistent with common quantitative finance practices. For each asset on each trading day, a feature vector $\mathbf{x} \in \mathbb{R}^6$ was constructed, comprising: 1) the standard log **Daily Return**; 2) the annualized **30-Day Rolling Volatility**; 3) a short-term **14-Day Momentum** factor; 4) a longer-term **60-Day Momentum** factor; 5) a **7-Day Short-Term Reversal** factor; and 6) the **30-Day Rolling Volume Average**, normalized by its yearly average. For the benchmark models, classical HRP and Markowitz optimization used standard daily returns as their input. All features were standardized before being input into the models. For the KHRP method, the Gaussian kernel bandwidth parameter, σ, was set to 1.0 during the walk-forward validation, as specified in the code.

4.2 Quantum Circuit Ansatz

In our approach, classical financial data is embedded into a quantum Hilbert space by means of a *quantum feature map*. In this section, we detail the design of the quantum circuit that implements this feature map, the encoding of the classical data into a quantum state, and the subsequent construction of the corresponding density matrix. Let $\mathbf{x} \in \mathbb{R}^P$ be a classical feature vector corresponding to one observation of an asset. Our goal is to map $\mathbf{x}$ into a quantum state $|\psi(\mathbf{x})\rangle$ in a Hilbert space of dimension $D = 2^n$, where n is the number of qubits used. This is accomplished by defining a parameterized unitary operator $U(\mathbf{x})$ such that:

$$|\psi(\mathbf{x})\rangle = U(\mathbf{x})\,|0\rangle^{\otimes n}\,, \tag{18}$$

and the corresponding density matrix is given by:

$$\rho(\mathbf{x}) = |\psi(\mathbf{x})\rangle\,\langle\psi(\mathbf{x})| = U(\mathbf{x})\,|0\rangle\,\langle 0|\,U(\mathbf{x})^{\dagger}. \tag{19}$$

This mapping $\varphi : \mathbf{x} \mapsto \rho(\mathbf{x})$ allows us to represent classical data in a quantum-enhanced feature space.

Consider an $n \times 1$ random vector X partitioned into K blocks, denoted as:

$$X = \begin{bmatrix} X^{(1)} \\ \vdots \\ X^{(K)} \end{bmatrix}, \quad X^{(k)} \in \mathbb{R}^{n_k}, \quad \sum_{k=1}^{K} n_k = n. \tag{20}$$

Here, each $X^{(k)}$ represents the subset of variables belonging to the k-th block, with block sizes n_k. We assume the presence of a global factor $F^{(0)} \in \mathbb{R}^{r_0}$ that influences all variables, as well as block-specific factors $F^{(k)} \in \mathbb{R}^{r_k}$ that affect only the corresponding block $X^{(k)}$. The covariance matrix Σ follows the decomposition:

$$\Sigma = \Lambda^{(0)}\Omega^{(0)}\Lambda^{(0)\prime} + \sum_{k=1}^{K} \Lambda^{(k)}\Omega^{(k)}\Lambda^{(k)\prime} + D. \tag{21}$$

In this formulation, $\Lambda^{(0)} \in \mathbb{R}^{n\times r_0}$ is the loading matrix for the global factors, $\Lambda^{(k)} \in \mathbb{R}^{n\times r_k}$ is the loading matrix for the block-specific factors, and $\Omega^{(0)} \in \mathbb{R}^{r_0\times r_0}$ and $\Omega^{(k)} \in \mathbb{R}^{r_k\times r_k}$ are the positive-definite covariance matrices of the global and block-specific factors, respectively. The matrix D is a block-diagonal residual structure given by:

$$D_{[k,k]} = (d_k - b_{kk})I_{n_k} + b_{kk}\mathbf{1}_{n_k}\mathbf{1}_{n_k}'. \tag{22}$$

Here, d_k represents the diagonal elements, and b_{kk} is the within-block constant correlation term, ensuring each block retains the equicorrelation structure observed in traditional block models.

In our implementation, we adopt *amplitude encoding* to embed the classical vector $\mathbf{x} = (x_0, x_1, \dots, x_{P-1})^\top$ into a quantum state. Since the Hilbert space is of dimension $D = 2^n$, we require that $P \leq D$. For encoding, we start with the n-qubit computational basis state

$$|0\rangle^{\otimes n} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \tag{23}$$

Then, we construct a D-dimensional vector $\tilde{\mathbf{x}}$ by placing the components of $\mathbf{x}$ in the first P positions and padding with zeros if $P < D$:

$$\tilde{\mathbf{x}} = \begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_{P-1} \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \tag{24}$$

Finally, we normalize $\tilde{\mathbf{x}}$ to obtain the state vector

$$|\psi(\mathbf{x})\rangle = \frac{\tilde{\mathbf{x}}}{\|\tilde{\mathbf{x}}\|}, \tag{25}$$

where $\|\tilde{\mathbf{x}}\| = \sqrt{\sum_{i=0}^{P-1} x_i^2}$.

In a physical quantum circuit, this procedure is implemented by constructing a unitary $U(\mathbf{x})$ that maps the initial state $|0\rangle^{\otimes n}$ to the desired state $|\psi(\mathbf{x})\rangle$. There exist various methods to realize amplitude encoding; one approach is to decompose $U(\mathbf{x})$ into a sequence of rotation gates and controlled operations such that

$$U(\mathbf{x}) = U_K(\mathbf{x}) \cdots U_2(\mathbf{x}) U_1(\mathbf{x}), \tag{26}$$

where each $U_k(\mathbf{x})$ is a parameterized gate (e.g., R_y rotations) whose rotation angles are functions of the components of $\mathbf{x}$. Although our classical simulation directly constructs the normalized state as in Equation (25), an actual quantum implementation would require the circuit in Equation (26).

The diagram of the quantum circuit ansatz is depicted in Figure 1. The n-qubit circuit is initialized in the all-zero quantum state $|0\rangle^{\otimes n}$. Then, a layer of parameterized one-qubit rotations around the x, y, and z axes $(R_x(\theta_i(\mathbf{x})), R_y(\theta_i(\mathbf{x})), R_z(\theta_i(\mathbf{x})))$ are applied to each qubit, followed by a layer of two-qubit gates (CNOT, CZ, SWAP, iSWAP, etc.) creating entanglement. Given our use of $P = 6$ features, the quantum circuit utilizes 6 qubits.

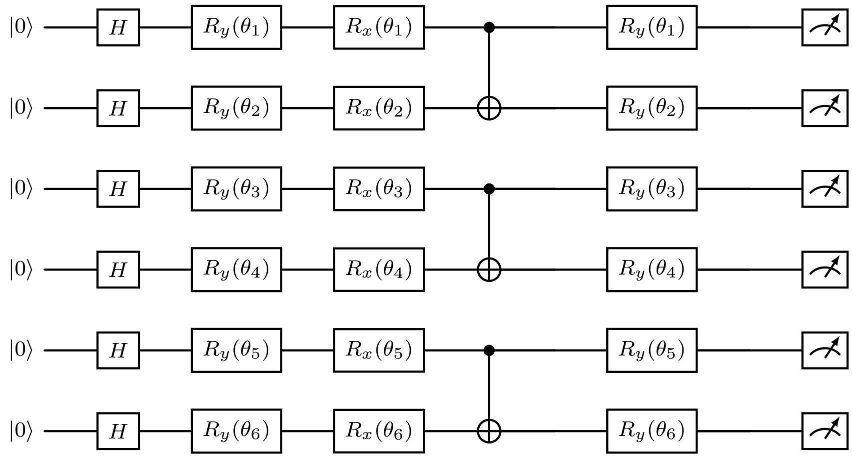


Fig. 1: Quantum circuit for encoding a 6-dimensional feature vector. The circuit includes: an initial Hadamard layer; three data encoding layers (using R_y and R_x rotations) with angles θ_i determined by normalized features scaled by α; and entangling gates (CNOT in this example) interleaved between encoding layers. Additional entanglement (using CZ, SWAP, or iSWAP) can be incorporated as needed.

The angles $\theta_i(\mathbf{x})$ are functions of the input features. For example, one may choose

$$\theta_i(\mathbf{x}) = \arctan\left(\frac{x_i}{\sqrt{\sum_{j \neq i} x_j^2}}\right), \quad i = 0, \ldots, P-1. \tag{27}$$

The layers of one-qubti and two-qubit gates realize the unitary $U(\mathbf{x})$, preparing the state $|\psi(\mathbf{x})\rangle$. Although our simulation directly computes the density matrix, on a quantum computer, the state $|\psi(\mathbf{x})\rangle$ would be measured or reconstructed via state tomography to yield $\rho(\mathbf{x})$.

In the proposed ansatz, the circuit uses an initial layer of Hadamrd gates to create a uniform superposition of basis states. In data encoding layers, each feature is embedded via rotations (R_y and R_x) whose angles are determined by normalized feature values. Finally, entangling layers (using CNOT, CZ, and SWAP gates) interleaved with the encoding layers distribute the information across qubits. The block diagram of the corresponding quantum feature map is shown in Figure 2.

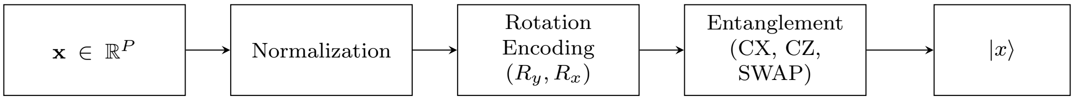


Fig. 2: Block diagram of the quantum feature map. The parameter α scales the rotation angles that encode each normalized feature.

A key hyperparameter in our encoding process is α, a dimensionless scalar that scales the rotation angles. Specifically, if x_i is the i-th feature of the input vector, and $\min_i, \max_i$ denote the minimum and maximum values of that feature in the training dataset, then we map x_i to a rotation angle

$$\theta_i \; = \; \alpha\,\pi\,\frac{x_i - \min_i}{\max_i - \min_i}. \tag{28}$$

Larger values of α produce larger rotation angles for a given feature range, potentially amplifying differences between data points and risking the circuit's saturation. Conversely, smaller α values yield subtler rotations, preserving finer distinctions in data at the cost of a weaker spread in the Hilbert space. In practice, α can be treated as a hyperparameter to tune or chosen based on heuristic considerations.

For a P-dimensional feature vector, we use P qubits. The circuit comprises three data encoding layers interleaved with entangling layers. Figure 1 illustrates the circuit (for $P = 6$).

4.3 Results

The following tables present the key out-of-sample results from our numerical experiments, which are the most indicative of real-world performance.

Table 1: Walk-Forward Out-of-Sample Performance Metrics

Method	**Mean Sharpe**	**Std of Sharpe**	**Mean PSR**
Quantum HRP	1.4658	3.7234	0.6172
Classical HRP	1.2190	3.8598	0.5970
Kernel-based HRP	1.2944	3.6671	0.6055
Markowitz	1.3474	3.7785	0.6093
Equal Weights	1.3565	3.9065	0.6043

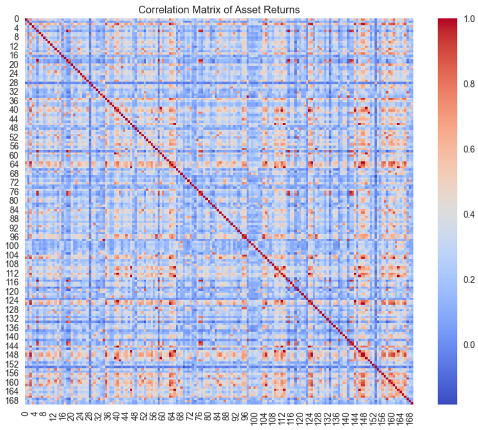
Table 2: Overall Out-of-Sample Sharpe Ratios from Aggregated Test Returns

Method	**Aggregated Out-of-Sample Sharpe**
Quantum HRP	1.3975
Classical HRP	1.1378
Kernel-based HRP	1.2186
Markowitz	1.2557
Equal Weights	1.1983

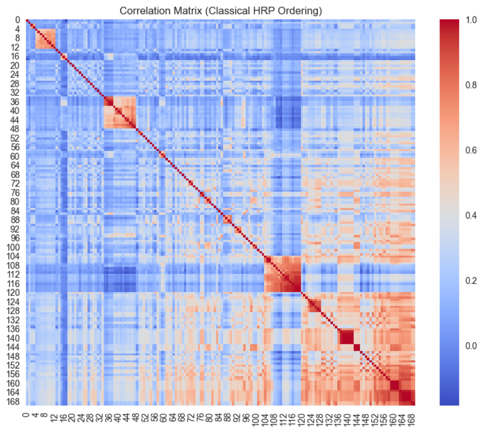
The walk-forward analysis (Table 1) and aggregated returns (Table 2) provide a robust evaluation of the models.

The walk-forward results are particularly compelling. Quantum HRP achieves the highest Mean Sharpe Ratio (1.4658), significantly outperforming all other methods, including its multi-feature classical counterpart, KHRP (1.2944). This suggests that the quantum feature mapping captures more predictive relationships than the classical kernel method. Interestingly, KHRP displays the lowest standard deviation of Sharpe ratios (3.6671), indicating that multi-feature HRP methods (both quantum and classical) may offer more performance stability than traditional approaches.

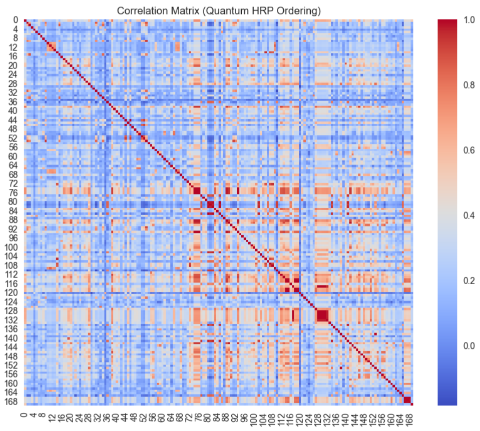
This narrative is confirmed by the aggregated out-of-sample Sharpe ratios (Table 2), where QHRP again ranks first with a Sharpe of 1.3975. KHRP (1.2186) provides a clear improvement over classical HRP (1.1378), confirming the value of using multiple features. However, its inability to match the Markowitz or QHRP results suggests that the quantum embedding provides a distinct advantage. The superior performance of QHRP in these rigorous out-of-sample tests underscores its potential to build more robust and adaptive portfolios.



(a) Correlation Matrix (Unordered)



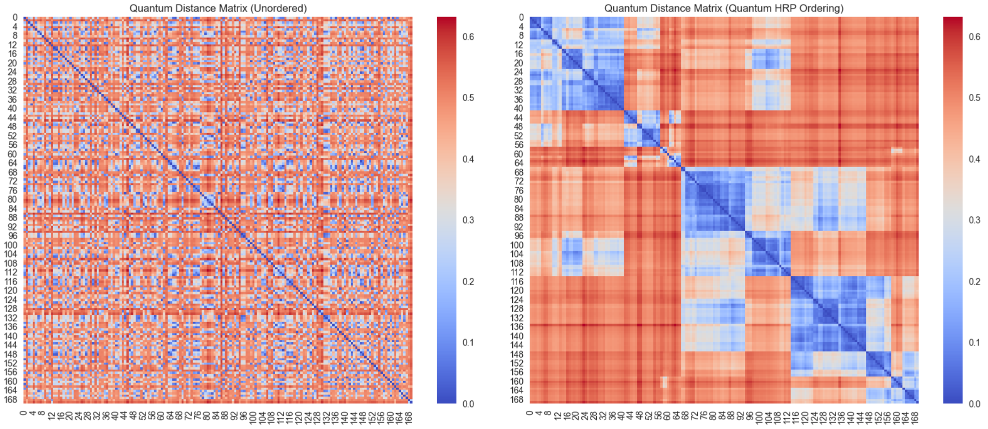
(b) Correlation Matrix (Classical HRP Ordering)



(c) Correlation Matrix (Quantum HRP Ordering)

Fig. 3: Correlation Matrices of Asset Returns Under Different Orderings

Figure 3 visually demonstrates the effect of the hierarchical clustering at the core of the HRP methodology. The unordered correlation matrix (a) shows a complex web of correlations without a clear structure. In contrast, the matrices ordered by Classical HRP (b) and Quantum HRP (c) are rearranged to be quasi-diagonal. This ordering places assets with high correlations (or, in the case of QHRP, high similarity in the quantum feature space) adjacent to one another, revealing distinct clusters along the main diagonal. Figure 4 provides direct insight into the asset relationships as captured by the quantum feature map and the Frobenius distance metric. The unordered matrix (a) shows the raw pairwise distances. After applying the tree clustering and reordering the assets, the resulting matrix (b) exhibits a



(a) Quantum Distance Matrix (Unordered) (b) Quantum Distance Matrix (Quantum HRP Ordering)

Fig. 4: Quantum Distance Matrices With and Without Ordering

distinct quasi-diagonal structure. The blue squares along the diagonal represent clusters of assets that are "close" to one another in the Hilbert space. This visualization confirms that the QHRP algorithm successfully partitions assets into coherent groups based on their quantum representations, which is essential for the effectiveness of the recursive bisection weight allocation stage.

5 Discussions

The development of Quantum Hierarchical Risk Parity (QHRP) represents a novel intersection of quantum computing and advanced portfolio management. The empirical results, drawn from a large-scale experiment with 170 unique assets, indicate that QHRP is not merely a theoretical curiosity but a practical methodology capable of delivering superior and more robust investment portfolios. This section contextualizes these updated findings, explores their implications, and outlines the limitations and future directions of this research.

5.1 Advancing Beyond Correlation-Based Risk Parity

A core limitation of traditional portfolio methods is their foundational reliance on the correlation matrix, a metric that is notoriously unstable and often fails during financial crises when diversification is most needed. Our empirical results demonstrate that QHRP successfully mitigates this dependency. By using a quantum feature map to encode a richer, six-dimensional feature set for each asset, QHRP moves beyond a single, often misleading, metric.

The out-of-sample performance provides strong evidence for this advancement. In the rigorous walk-forward analysis, QHRP achieved a mean Sharpe ratio of 1.4658, and an aggregated out-of-sample Sharpe ratio of 1.3975 across the entire test period. This was the highest performance achieved among all tested methods, indicating that the quantum approach captures more durable and predictive relationships between assets than classical methods. This aligns with the central thesis that leveraging more complex feature sets is essential for navigating modern financial markets.

5.2 The Power of Quantum vs. Kernel-Based Feature Maps

This study introduced both a quantum (QHRP) and a classical kernel-based (KHRP) method to overcome the single-feature limitation of the original HRP algorithm. The inclusion of KHRP as a “fair classical counterpart” allows for a direct comparison between advanced classical and quantum techniques.

The results show that both methods improve upon the original HRP. KHRP, using a Gaussian kernel, achieved an aggregated Sharpe ratio of 1.2186, significantly outperforming classical HRP’s 1.1378. This confirms that simply using more features, even with a classical non-linear method, is beneficial. However, QHRP’s aggregated Sharpe ratio of 1.3975 demonstrates a substantial performance leap over KHRP. This suggests that the quantum feature mapping, which uses entanglement to model interdependencies, provides a more powerful and effective way to represent asset data and distinguish between their behaviors than the classical kernel approach. The Frobenius distance between quantum states appears to be a more discerning similarity measure in this context.

5.3 Limitations and Future Research

While the results are promising, it is crucial to acknowledge the limitations of this study and identify avenues for future work.

First, the experiments in this paper were conducted on classical simulators of quantum computers. The performance of the QHRP algorithm on current, noisy intermediate-scale quantum (NISQ) devices remains an open question. Future research should focus on implementing and testing QHRP on physical quantum hardware to validate its real-world viability and resilience to quantum noise, a critical step towards practical quantum advantage in finance.

Second, this study used a fixed set of six features and set hyperparameters like the quantum scaling factor ($\alpha = 2.0$) and kernel bandwidth ($\sigma = 1.0$) based on the implementation code. A promising direction for future work is to perform a systematic study of feature importance and conduct a rigorous hyperparameter optimization to potentially unlock further performance gains.

Finally, the analysis was performed on a large portfolio of 170 assets. While impressive, institutional portfolios can be even larger. Investigating the scalability of the QHRP algorithm is therefore paramount. This is an area where fault-tolerant quantum computers are expected to offer a significant speedup over classical machines, making QHRP a forward-looking solution for large-scale financial optimization.

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A Constituent Assets of Our Diversified Portfolio

The following table details the complete list of constituent assets used in the numerical experiments. The list contains 170 assets from various classes worldwide.

Table 3: List of Constituent Assets

Ticker	**Category**	**Ticker**	**Category**
	Major US Index ETFs		
SPY	Major US Index ETFs	QQQ	Major US Index ETFs
DIA	Major US Index ETFs	IWM	Major US Index ETFs
IVV	Major US Index ETFs	VTI	Major US Index ETFs
VOO	Major US Index ETFs		
	Global Index ETFs		
EFA	Global Index ETFs	EEM	Global Index ETFs
VEA	Global Index ETFs	VWO	Global Index ETFs
ACWI	Global Index ETFs	SPDW	Global Index ETFs
SCHF	Global Index ETFs		
	Sector ETFs		
XLK	Sector ETFs	XLF	Sector ETFs
XLE	Sector ETFs	XLI	Sector ETFs
XLY	Sector ETFs	XLV	Sector ETFs
XLC	Sector ETFs	XLP	Sector ETFs
XLRE	Sector ETFs	XLU	Sector ETFs
XLB	Sector ETFs		
	Real Estate & REITs		
VNQ	Real Estate & REITs	IYR	Real Estate & REITs
RWX	Real Estate & REITs	SCHH	Real Estate & REITs
FREL	Real Estate & REITs	RWO	Real Estate & REITs
	Intl. & EM Bonds		
BNDX	Intl. & EM Bonds	VWOB	Intl. & EM Bonds
IGOV	Intl. & EM Bonds	IEMG	Intl. & EM Bonds
EMB	Intl. & EM Bonds		

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Table 3 – continued from previous page

Ticker	**Category**	**Ticker**	**Category**
	Commodities		
GLD	Commodities	SLV	Commodities
PPLT	Commodities	COPX	Commodities
DBB	Commodities	DBC	Commodities
PDBC	Commodities	USO	Commodities
UGA	Commodities	CORN	Commodities
WEAT	Commodities	SOYB	Commodities
	Bonds & Fixed Income		
TLT	Bonds & Fixed Income	IEF	Bonds & Fixed Income
LQD	Bonds & Fixed Income	HYG	Bonds & Fixed Income
SHY	Bonds & Fixed Income	TIP	Bonds & Fixed Income
AGG	Bonds & Fixed Income	BND	Bonds & Fixed Income
MBB	Bonds & Fixed Income	SCHZ	Bonds & Fixed Income
VCSH	Bonds & Fixed Income	VCIT	Bonds & Fixed Income
	Crypto		
BTC-USD	Crypto	ETH-USD	Crypto
SOL-USD	Crypto	BNB-USD	Crypto
XRP-USD	Crypto	ADA-USD	Crypto
DOGE-USD	Crypto	DOT-USD	Crypto
AVAX-USD	Crypto		
	Large-Cap US Stocks		
AAPL	Large-Cap US Stocks	MSFT	Large-Cap US Stocks
GOOGL	Large-Cap US Stocks	AMZN	Large-Cap US Stocks
NVDA	Large-Cap US Stocks	TSLA	Large-Cap US Stocks
BRK-B	Large-Cap US Stocks	META	Large-Cap US Stocks
NFLX	Large-Cap US Stocks	DIS	Large-Cap US Stocks
	Financials		
JPM	Financials	GS	Financials
BAC	Financials	C	Financials
WFC	Financials	MS	Financials
AXP	Financials	BLK	Financials
SCHW	Financials	TFC	Financials
	Industrials & Transportation		
BA	Industrials & Transportation	CAT	Industrials & Transportation
HON	Industrials & Transportation	LMT	Industrials & Transportation
UNP	Industrials & Transportation	CSX	Industrials & Transportation
NSC	Industrials & Transportation	UPS	Industrials & Transportation
FDX	Industrials & Transportation	RTX	Industrials & Transportation
	Energy		
CVX	Energy	XOM	Energy
OXY	Energy	MPC	Energy
PSX	Energy	VLO	Energy
SLB	Energy	HAL	Energy
BKR	Energy		
	Consumer Staples & Discretionary		
PG	Consumer Staples & Discretionary	KO	Consumer Staples & Discretionary
PEP	Consumer Staples & Discretionary	MO	Consumer Staples & Discretionary
PM	Consumer Staples & Discretionary	WMT	Consumer Staples & Discretionary
TGT	Consumer Staples & Discretionary	HD	Consumer Staples & Discretionary
LOW	Consumer Staples & Discretionary	MCD	Consumer Staples & Discretionary
SBUX	Consumer Staples & Discretionary	NKE	Consumer Staples & Discretionary
TJX	Consumer Staples & Discretionary		

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Table 3 – continued from previous page

Ticker	**Category**	**Ticker**	**Category**
	Healthcare & Pharma		
JNJ	Healthcare & Pharma	PFE	Healthcare & Pharma
UNH	Healthcare & Pharma	CVS	Healthcare & Pharma
ABBV	Healthcare & Pharma	LLY	Healthcare & Pharma
MRNA	Healthcare & Pharma	REGN	Healthcare & Pharma
BMY	Healthcare & Pharma	GILD	Healthcare & Pharma
	Tech, AI & Innovation		
AMD	Tech, AI & Innovation	INTC	Tech, AI & Innovation
TSM	Tech, AI & Innovation	ORCL	Tech, AI & Innovation
IBM	Tech, AI & Innovation	ADBE	Tech, AI & Innovation
CRM	Tech, AI & Innovation	SNOW	Tech, AI & Innovation
PANW	Tech, AI & Innovation	AVGO	Tech, AI & Innovation
	Communication & Media		
VZ	Communication & Media	T	Communication & Media
CMCSA	Communication & Media	CHTR	Communication & Media
TMUS	Communication & Media	WBD	Communication & Media
	International Stocks		
BABA	International Stocks	TCEHY	International Stocks
INFY	International Stocks	SHOP	International Stocks
SE	International Stocks	MELI	International Stocks
BYDDF	International Stocks	NIO	International Stocks
	Auto Industry		
GM	Auto Industry	F	Auto Industry
RIVN	Auto Industry	LCID	Auto Industry
NKLA	Auto Industry	XPEV	Auto Industry
LI	Auto Industry		
	ESG & Green Energy		
ICLN	ESG & Green Energy	TAN	ESG & Green Energy
PBW	ESG & Green Energy	QCLN	ESG & Green Energy
LIT	ESG & Green Energy	FAN	ESG & Green Energy
PBD	ESG & Green Energy	KRBN	ESG & Green Energy